

ST-ETIENNE BOUTHEON AP, France

WMO: 074750

Lat: 45.5331N

Lon: 4.2933E

Elev: 1325

StdP: 14.01

Time Zone: 1.00 (EUC)

Period: 99-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	19.5	23.0	13.1	11.2	22.8	17.3	13.8	26.3	21.1	49.5	18.8	47.7	4.4	330	0.490

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	22.6	91.7	67.4	87.8	66.6	84.3	65.6	69.8	85.6	68.2	83.4	66.8	80.9	10.0	170

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
64.7	96.1	75.0	63.2	91.2	73.3	61.9	87.0	72.1	34.5	85.9	33.2	83.4	32.0	81.2	77.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a) 20.0	(b) 16.7	(c) 14.4	(e) 13.7	(f) 97.0	(g) 5.0	(h) 3.5	(i) 10.0	(j) 99.5	(k) 7.1	(l) 101.6	(m) 4.3	(n) 103.5	(o) 0.6	(p) 106.1		
			DB	WB	DB	WB	DB	WB	DB	WB	DB	WB	DB	WB		
			12.6	72.8	4.9	1.7	9.0	74.0	6.1	75.0	3.4	75.9	-0.2	77.2		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	53.2	38.2	39.5	45.6	51.2	58.0	65.5	68.9	68.0	61.7	55.0	45.5	39.9
	DBStd	13.04	7.77	8.11	7.09	6.40	6.88	7.15	6.39	6.50	6.36	7.38	7.58	7.84
	HDD50	1449	372	299	170	61	10	0	0	2	38	174	323	
	HDD65	4764	831	714	600	414	236	80	32	41	136	315	585	780
	CDD50	2605	6	5	35	98	258	466	586	557	354	193	39	8
	CDD65	444	0	0	0	1	18	96	153	133	38	5	0	0
	CDH74	4966	0	0	1	33	236	1085	1684	1479	413	35	0	0
	CDH80	1842	0	0	0	1	50	408	668	616	97	2	0	0
(14) Wind	WSAvg	5.8	5.8	6.2	6.4	6.5	6.1	5.7	5.5	5.2	5.0	5.8	6.0	5.9
(15) Precipitation	PrecAvg	29.3	1.7	1.4	1.3	2.1	3.0	3.0	3.0	2.7	2.9	3.2	3.3	1.8
	PrecMax	37.9	3.7	2.7	4.1	5.5	5.3	7.2	7.0	6.7	7.6	6.3	8.3	6.6
	PrecMin	20.2	0.2	0.3	0.0	0.2	0.7	0.9	1.0	1.0	0.6	0.4	1.4	0.4
	PrecStd	4.5	0.9	0.7	0.9	1.2	1.3	1.3	1.3	1.2	1.7	1.5	1.7	1.2
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	60.3	62.8	71.0	78.5	85.6	94.8	96.2	97.2	87.2	78.4	67.6	60.7
		MCWB	51.3	49.4	53.1	58.2	64.2	68.0	67.8	67.2	65.0	61.5	56.8	51.9
	2%	DB	56.2	58.1	65.9	72.9	80.3	89.1	91.1	91.7	83.0	73.7	63.3	57.4
		MCWB	49.0	48.2	51.3	56.6	62.2	67.2	67.7	67.3	64.1	60.5	54.5	49.8
	5%	DB	53.0	54.9	62.1	68.6	75.8	84.6	87.1	87.0	79.0	70.2	60.5	55.1
		MCWB	46.8	46.6	50.4	54.1	60.2	66.0	66.7	66.4	62.9	59.6	52.8	48.6
	10%	DB	50.1	52.0	58.3	64.6	71.8	80.6	83.4	82.5	75.0	67.2	57.6	52.1
		MCWB	44.7	44.8	48.8	52.7	58.4	64.6	65.6	65.4	61.5	58.1	51.3	46.6
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	51.9	51.0	55.6	60.4	67.1	71.7	71.7	71.0	68.3	64.2	57.5	52.9
		MCDB	58.7	59.4	66.7	75.4	79.8	88.0	87.8	87.2	81.1	74.5	65.6	59.3
	2%	WB	49.5	48.9	52.9	57.5	63.9	69.3	69.7	69.1	65.7	62.1	55.3	50.6
		MCDB	55.5	56.6	62.8	69.7	76.7	84.2	85.5	85.5	78.9	71.1	61.7	56.3
	5%	WB	47.3	47.2	51.1	55.4	61.7	67.3	68.1	67.5	63.9	60.3	53.5	49.0
		MCDB	52.5	54.2	60.7	66.8	72.8	80.8	83.6	82.9	75.4	68.6	59.8	54.3
	10%	WB	45.1	45.3	49.3	53.2	59.6	65.5	66.6	66.0	62.4	58.5	51.6	47.1
		MCDB	49.6	51.3	57.7	62.9	69.4	77.9	80.5	80.0	73.0	65.8	57.1	51.8
(35) Mean Daily Temperature Range	5% DB	MDBR	12.3	15.0	18.7	20.1	20.3	22.0	22.6	22.9	20.9	17.2	13.6	12.6
		MCDBR	16.2	21.6	25.3	28.1	26.6	28.5	29.0	30.6	27.5	21.8	18.6	16.4
		MCWBR	10.8	13.6	14.3	14.5	12.6	12.3	11.5	11.8	12.2	11.5	11.5	10.7
	5% WB	MCDBR	15.4	19.9	22.8	26.1	24.4	26.2	26.6	27.1	23.5	20.0	17.0	15.3
		MCWBR	10.5	12.9	13.2	14.1	12.5	12.0	11.6	11.9	11.3	11.0	10.8	10.3
(40) Clear-Sky Solar Irradiance	taub		0.301	0.322	0.351	0.388	0.393	0.396	0.392	0.387	0.366	0.354	0.320	0.298
	taud		2.485	2.425	2.364	2.267	2.304	2.329	2.336	2.359	2.398	2.434	2.481	2.493
	Ebn at Noon		258	272	278	276	278	277	276	273	270	256	247	246
	Edh at Noon		22	27	33	40	40	39	38	36	32	27	22	19
(44) All-Sky Solar Radiation	RadAvg		406	677	1082	1428	1667	1983	1954	1692	1294	815	456	344
	RadStd		48	94	130	181	158	157	164	127	99	81	53	58

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (33 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page